

ANIMA Phase I: Emergent Organization and the Architectural Limits of Persistent Learning in a Minimally Specified Spiking Nervous System

ANIMA project
Experimental developmental artificial nervous system — Phase I record

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Abstract

ANIMA is an experimental developmental artificial nervous system: a deterministically seeded spiking substrate in which only the boundaries — sensory input, output, environment, curriculum, resource limits, and local plasticity laws — are specified, and nothing else. Phase I asked whether useful, stable, adaptive internal organization can arise from such a minimal specification, and progressed through a series of frozen, identity-gated experiments to locate the architectural limit of persistent learning. The record demonstrates, in turn: sensory-specific synaptic organization emerges from spike-timing-dependent plasticity under purely local rules; transient spike-based representations exist during stimulation; scalar intrinsic persistence is information-poor and cannot serve as sensory memory; the synaptic weight structure is the substrate’s persistent store, but it suffers order-dependent interference — blocked sequential experience overwrites, alternated experience superposes; bounded consolidation protects the store from runaway and preserves alternating coexistence; and, once stability was established, the limiting factor moved successively to allocation, to first-exposure allocation, and finally to the recruitment of postsynaptic activity for genuinely novel late-arriving patterns, which Phase I leaves unresolved. Throughout, the project enforced pre-registered protocols, byte-identical flag-off control gates, deterministic replay, and preservation of negative results, so that each negative narrows the space of possible architectures rather than being discarded. The paper reports what is demonstrated, what is supported, what is unmeasured, and what remains hypothesis, with every quantitative claim traceable to a committed repository record.

Keywords: emergent organization; spiking neural substrate; persistent synaptic structure; synaptic interference; allocation; identity discipline; pre-registration; minimal specification.

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1 Introduction

Biological nervous systems are not engineered as circuits; they develop. A research program that takes development seriously therefore faces a choice of specification: prescribe the final organization (a designed network), or prescribe only the materials and the laws and ask whether organization arises. The ANIMA project takes the second path. Its charter fixes only the sensory boundary, the output boundary, the environment and curriculum, finite resource limits, and fundamental local plasticity laws, and requires that the organism — a population of spiking neurons — develop useful, stable, adaptive internal organization through interaction, without any prescription of the final organization [19, 7].

Three commitments follow from this framing and structure everything reported here. First, *minimal specification*: no task-specific semantic variables, no labels, no reward or credit assignment, no hidden supervision, and no hand-designed internal architecture. The organism’s “concepts” are whatever its plasticity laws and its curriculum make of its inputs. Second, *locality*: all mechanisms operate on local state — membrane potentials, arriving currents, synaptic weights, and per-neuron budgets. Third, *determinism and discipline*: every experiment is frozen in writing before implementation, runs under a single seeded random number generator so that same-seed reproduction is byte-identical, is gated by a flag-off identity check against the committed baseline, and preserves its negative results [19].

The research question of Phase I is deliberately modest in form and demanding in content:

In a minimally specified spiking substrate with local plasticity and finite resources, what internal organization emerges, what of it persists, and what are the architectural limits of persistent, retrievable learning under sequential experience?

The answer, as measured rather than assumed, is summarized by the progression in Figure 1 and developed through the rest of the paper. The phrase “architectural limits” is used advisedly: Phase I’s contribution is not a system that works, but a map of the causal chain through which each candidate substrate failed, with the failure point identified at each step. The final step — recruiting postsynaptic activity for a genuinely novel pattern that arrives late in a blocked curriculum — is a measurement of where the current architecture stops, not a claim that the class of architectures is exhausted.

1.1 Relation to related work

The substrate is an orthodox leaky-integrate-and-fire (LIF) population with pairwise additive spike-timing-dependent plasticity (STDP) [23, 25]. The aspiration — that competitive Hebbian learning organizes selective assemblies — has a long theoretical lineage [24, 26]. ANIMA differs from that lineage in its discipline rather than its ingredients: organization is never scored by a task the experimenter cares about; the curriculum is a fixed, deterministic schedule of synthetic pattern presentations; and every mechanism claim is tested under flag-off byte-identity control. The value of the record is therefore mostly negative: it shows, under these controls, exactly which attractive properties of the ingredients fail to materialize, and where successive local amendments move the failure.

2 Design Philosophy

The operating constraints that shaped every experiment are documented in the charter and decision log [19, 7]. They are restated here because they are load-bearing for the interpretation of every result.

Specified interfaces, not architectures. The project defines the sensory front end (input channels with deterministic Poisson spike trains per presentation), the output boundary (ordinary plastic neurons whose spikes are additionally emitted as output activity — a role tag, not a cell type), the environment/curriculum (an ordered schedule of pattern presentations), resource limits (neuron, synapse, per-neuron budget caps), and the plasticity laws (STDP, weight normalization, candidate/prune machinery, anti-Hebbian inhibition). It never defines assemblies, readout circuits, or any intermediate organization.

Local rules. Every mechanism is required to read only per-neuron or per-synapse state. This is enforced loosely in design and strictly at review: the CLLA falsifier F5 requires static conformance — consolidation and allocation decision paths may not read pattern identity, channel-group membership tests, presentation counts, or any global state [1, 6].

Finite resources. A neuron owns a total excitatory budget $t_e = 0.8$ (weight units, M2 normalization) and a candidate pool of six slots; consolidation may protect at most $p_{\max}t_e = 0.60$ of the budget; budgets bind synapse creation (M5, $b_e = 40$ excitatory afferents per neuron). Resource limits are treated as part of the specification, not as an implementation nuisance.

No semantic supervision. All presented patterns are examples; the curriculum is a configurable schedule, not a set of preinstalled concepts. The organism is never told which pattern it is seeing, whether a response was correct, or that a pattern is “novel.”

Reproducibility and freeze discipline. Protocols are committed before implementation; amendments are logged and user-approved; every execution is byte-deterministic under a seeded Xoshiro generator; a flag-off “identity gate” must reproduce the committed baseline byte-for-byte (event-stream hash and snapshot frames) before any flag-on run is interpreted; and negative results are preserved with their runs [19, 1]. Several protocol defects were discovered only by this discipline and are reported as such (§10, §4).

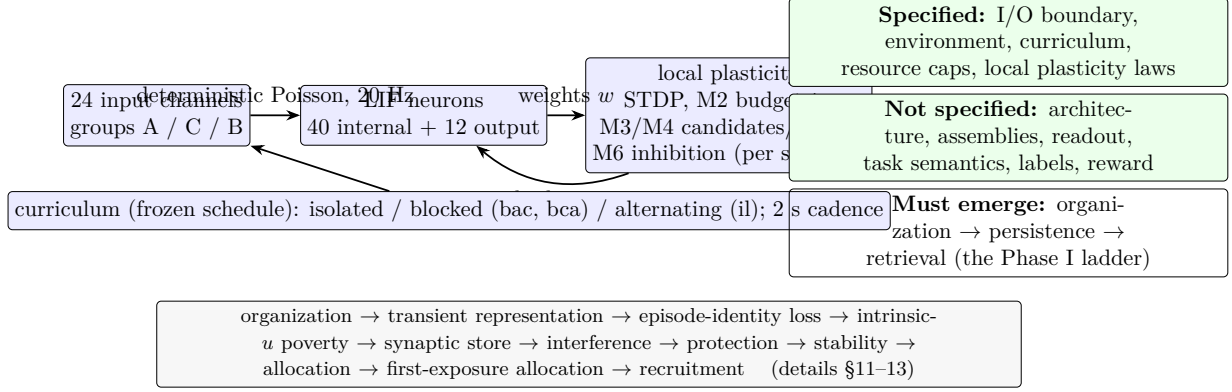


Figure 1: **ANIMA architecture and experimental philosophy.** Top: the specified boundary — 24 input channels (groups A, C, B), 64 LIF neurons (40 internal, 12 output plus input neurons), sparse seeded wiring, local plasticity laws, deterministic curriculum. Bottom left: the charter— what is specified vs. what must emerge. Bottom right: the phase progression — the causal narrowing of the persistent- learning failure reported in this paper (detail in §14).

3 The ANIMA Substrate

Table 1 lists the frozen substrate parameters used throughout Phase I; the authoritative spec is the V2 protocol [2, 19].

3.1 Neuron and network dynamics

Non-input neurons integrate

$$\frac{dv_i}{dt} = \frac{-(v_i - v_{\text{rest}}) + i_{\text{syn},i} + i_{\text{ext},i} - i_{\text{adapt},i} + u_i}{\tau_m}, \quad (1)$$

discretized at 1 ms with exponential kernel decay for i_{syn} (τ_{syn}), adaptation i_{adapt} (E3-era U1; injection `adaptation_gain` = 0.05 per spike), and the slow depolarizing state u (V2.1),

$$u_i(t+1) = u_i(t) e^{-1/\tau_s} + \beta \cdot \mathbf{1}[i \text{ spiked at } t], \quad (2)$$

with $\beta = 0$ restoring byte-identical V2 behavior. A neuron spikes when $v_i \geq v_{\text{th}}$ after its refractory period; it resets v_i to v_{rest} and deposits $a \cdot w$ (excitatory) or $-a \cdot w$ (inhibitory) onto each postsynaptic target, with a one-tick transmission delay. Throughout Phase I the birth triggers were disabled (`birth_trigger` = "none"); all results concern the fixed population.

3.2 Sensory interface and curriculum

Each input channel produces deterministic Poisson trains (seeded per pattern, repetition, and channel): 8 channels fire at 20 Hz for 500 ms, followed by 1500 ms of silence. Patterns A , C , and B are channel groups; pattern $D = A \cup C$ (channels 0–15, same rate) is used as the composition probe. The curriculum is a configurable stage schedule; the CLLA-era corpus uses three orders of patterns over three seeds:

- `il`: A/C interleaved, 20 presentations each (40 total);

Table 1: **Substrate specification (frozen V2; the V2 protocol [2]).** All values commit in the committed configuration files; later experiments change only the listed flags.

Layer / mechanism	Value	Note
Tick	1 ms	deterministic; seeded
Input channels	24 (groups $A = 0..7$, $C = 8..15$, $B = 16..23$)	pure spike sources (D8): no internal state
Internal / output neurons	40 / 12 (ids 24..75, $N = 52$)	LIF; output is a role tag
LIF	$\tau_m = 20$ ms, $v_{\text{rest}} = 0$, $v_{\text{th}} = 1$, refractory 2 ms	
Synaptic current	$\tau_{\text{syn}} = 5$ ms, amplitude $a = 52$	current per spike = $a \cdot w$
Initial wiring	M1: input $p = 0.5$, recurrent $p = 0.2$, weak weights	seeded
STDP	pairwise additive; $\tau_{\pm} = 20$ ms, $a^+ = 0.005$, $a^- = 0.0053$	E1 amendment A1 operating point
Passive decay	10^{-6} /tick	
M2 normalization	per-neuron excitatory budget $t_e = 0.8$	multiplicative rescale of live incoming
M3 candidates	6 slots; $w_{c,\text{init}} = 0.01$, $\theta_{\text{perm}} = 0.05$, $w_{c,\text{perm}} = 0.02$	co-active windows
M4 pruning	$\theta_{\text{prune}} = 0.005$, 10 windows	excitatory only
M5 budgets	$b_e = 40$ exc / $b_i = 10$ inh per neuron	
M6 inhibition	anti-Hebbian; $w_{\text{inh},\text{max}} = 0.10$	
E6	per-channel rate balance, history-derived	no labels
V2.1 slow state	u : $+\beta/\text{spike}$, decay τ_s ; $\beta = 0.0046875$, $\tau_s = 5000$ ms	the e24 cell used by all CLLA-era runs
Resource detector	runaway: population mean > 50 Hz sustained 5 s	abort with failure record
Curriculum cadence	500 ms presentation @ 20 Hz/channel; 1500 ms off	2 s per presentation
Seeds	20260912, 424242, 9001	frozen

- **bac**: $20 \times A$, then $20 \times C$;
- **bca**: $20 \times C$, then $20 \times A$;
- **d**: $40 \times D$ (composition probe, informative only in most CLLA records).

3.3 Plasticity in normative form

STDP is the pairwise additive trace rule [25, 23], applied per tick as

$$\Delta w = a^+ \text{pre-trace} \cdot \mathbf{1}[\text{post fired}] - a^- \text{post-trace} \cdot \mathbf{1}[\text{pre fired}], \quad (3)$$

with trailing traces $\tau_{\pm} = 20$ ms and weights clamped to $[0, 1]$. Where the implemented equations deviate from the normative form (notably the dimensionality of the allocation gate’s drive estimate; §11, §19), the records are explicit, and the paper preserves the correction history.

4 Experimental Methodology

Phase I’s methodology is the paper’s main methodological contribution: a way of running negative experiments in which the negative is attributable. Five practices recur in every era.

Frozen protocols. A protocol document (configuration, arms, measurement definitions, falsifier inequalities, verdict rule) is committed before any implementation. Amendments are separate user-approved documents. The protocol text, not the result, governs the verdict; the CLLA bundle protocol [1] is the canonical example, with falsifiers F1–F7 and a binary verdict rule (§10).

Identity gates. Every mechanism is implemented behind a configuration flag whose default is off. Before any flag-on run is interpreted, a flag-off run must reproduce the committed baseline byte-for-byte: the event-stream FNV-1a hash and every snapshot frame. The CLLA-era identity anchor is the same-seed alternated baseline (152,254-row event stream, hash 9647ea8a0ca4dbd2), reproduced exactly by the unbounded, bounded, rule, reserve, first-exposure, and source-correction implementations [6, 4, 3, 8, 12, 15]. The recruitment-gain implementation additionally re-verified flag-off byte-identity against the corrected-fe baseline before its matrix [16]. This gate makes “the flag did nothing” a measured condition rather than a claim.

Deterministic replay and read-only audits. Every run is preserved under `runs/<id>/` with telemetry chunks, snapshots (1 s cadence), metrics, and report. Audits — the information-bottleneck audit [18], the V2.1 information audit [22], the weight-growth, drive, and recruitment audits [20, 14] — are read-only instruments over committed artifacts: they never re-run the organism. This lets later epochs re-examine earlier epochs’ claims at full resolution.

Falsifiers and preservation of failure. Experiments pre-register the inequality whose failure destroys the hypothesis (e.g., CLLA F1–F7; allocation S1 second-block/first-block mass ratio ≥ 0.5). Failures are preserved with their runs, including aborts: 6 of 12 unbounded-CLLA runs aborted on the runaway detector, and the record reports them rather than re-running the survivors [6]. Abort-loaded verdicts follow the frozen rule (“uninformative $\Rightarrow$ FAIL”).

Exploratory versus formal experiments. X-series and SDE-series probes run without E-numbers and cannot carry falsifier verdicts; formal E-numbered experiments and CLLA-era registered experiments carry the verdict language. Promotion requires an interesting, discriminating, reproducible result [19]. The recruitment-gain experiment (§13) is a registered experiment without an E-number, closed under its own outcome rules.

Record corrections preserved. Where the record itself was defective, the correction is part of the record, not a silent edit. Two are scientifically relevant and appear in this paper: the V2.1 Stage-C validation table was committed without executing any Stage-C runs; the information audit discovered that the referenced runs do not exist, retracted the verdict as ungrounded, and added an integrity regression test preventing phantom run references [22, 19]. And the CLLA capability protocol’s F2/F6 measurement windows (presentations 21–40 of each pattern) are empty by construction of the interleaved curriculum (20 presentations per pattern); the capability record reports F2/F6 as unmeasurable under the frozen definition rather than substituting a window, and reports the closest available window as a non-decisive diagnostic [5]. A third correction — the first-exposure allocator source vacuity — is handled in §11 because it changed the mechanism under test.

5 Emergent Sensory Organization

The first claim Phase I establishes is that the minimally specified substrate develops sensory-specific organization without any semantic supervision. Three levels of evidence exist in the record, in increasing order of specificity.

5.1 Afferent organization

Under repeated single-pattern training (40 presentations of pattern A), the afferent weight vectors become sharply selective. At drive end, the mean per-neuron mass is 0.290 for A -channel afferents against 0.026 for the never-presented C channels (mean selectivity $(A - C)/(A + C) = +0.906$, 49 of 52 neurons A -favoring). Training on C produces the mirror image (-0.758 , 2 of 52 A -favoring). The two trained configurations are mutually distant (full-matrix cosine 0.119; a leave-one-pair-out centroid decoder recovers the training identity in 91% of neurons) and stable across 20 seconds of post-drive silence (cosine to drive-end 0.986–0.993) [18]. Figure 2(a) shows the per-neuron afferent mass split for the A -trained run.

5.2 Transient spike-pattern separation

During the stimulus, the population’s response is organized but the pattern classes overlap. In E1’s curriculum (patterns $A/B/C$ interleaved, 360 presentations), within-pattern response profiles across S1 halves correlate 0.97–0.99 — stable assemblies — while cross-pattern profiles remain high ($A \cdot B$ 0.79→0.73, $A \cdot C$ 0.75→0.70, $B \cdot C$ 0.81→0.81 early to late S1) [9]. Later separation attempts quantified the plateau: mean late-S1 cross-pattern cosine 0.673 (E3) and 0.740 (E3b, with lateral inhibition), against a frozen 0.60 bar; median selectivity 0.455 and 0.366 against a 0.50 bar [11]. Both dynamics interventions failed in the same direction, which the records interpret as a representational (shared recurrent pool, no inhibition-derived gating), not a rate-dynamics, limit.

5.3 Per-presentation count structure

Under the CLLA-era curriculum the response is a per-presentation count structure: 500 ms presentations of a learned pattern recruit nearly the whole population with several thousand internal spikes (e.g. the A -block presentations of the baseline blocked run fire 1,300–1,700 internal spikes each), collapsed to a 52-dimensional spike-count vector. The count vector is the representation the readout instruments use, and the one the information bottleneck dilutes (§6); Figure 2(b) shows the per-presentation counts across the first block.

Taken together the records support:

Demonstrated: sensory-specific afferent structure emerges from local STDP, budget normalization, and pruning without supervision; per-presentation spike-count vectors are stimulus-locked and repeatable.

Supported: stable per-pattern assemblies exist, but their cross-pattern separation is weak and resisted two dynamics interventions (E3 adaptation, E3b inhibition).

Not established: any claim about separation sufficient for pattern decoding from population activity alone at a scientifically useful threshold (the E3b bar failed).

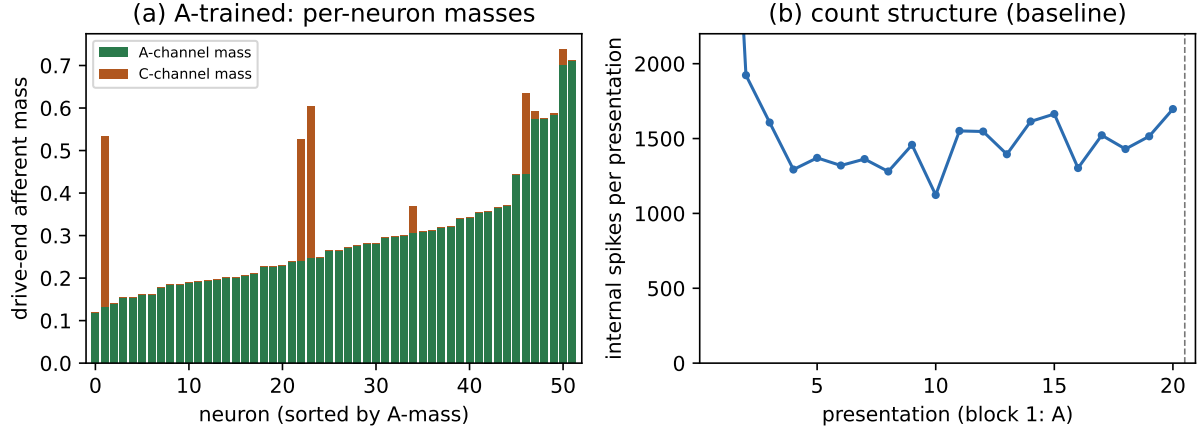


Figure 2: **Emergent sensory organization.** (a) Drive-end per-neuron afferent mass split (A-channel vs. C-channel) for the A-trained single-pattern run; the population organizes along the presented channel group (selectivity $+0.906$, 49/52 neurons) [18]. (b) Per-presentation internal spike counts over the first (A) block of the baseline blocked run — the repeatable count structure that subsequent regimes inherit. Data from preserved runs via the committed instrument `rg8eval`.

6 The Persistent-State Information Bottleneck

The central diagnostic of Phase I is an information-loss ladder. Consider what the environment provides and what the persistent variables of the substrate can possibly retain. Figure 3 schematizes the cascade; each rung is a compression the substrate performs automatically, and each was measured where the instruments allowed.

Rung 1: afferent structure. Patterns are channel groups; the persistent synaptic structure develops selectivity for them (selectivity ± 0.8 – 0.9 , §5). At this rung, structure exists and survives silence.

Rung 2: instantaneous spikes. At any instant, the internally observed quantity is the spike train. Spikes carry the full timing information of the stimulus, but they are activity: the substrate does not store them.

Rung 3: windowed counts. The measured representation of a presentation is its 500 ms spike-count vector. This is the rung at which the organization is assayed (response separation) and it is stimulus-locked: off-window (count-free) responses do not separate. The architectural review measured during-window cross-pattern separation cosines 0.30 – 0.89 but off-window $0.999+$, and the V2.2-era and E21 records fixed the temporal carry of the substrate below 50 ms (gap-0 retention only: $D_{L-NF} = 0.0815$) [19].

Rung 4: cumulative counts. The substrate has one integrator that accumulates over presentations: the slow state u (V2.1), $u \pm \beta$ per spike with decay τ_s . Under drive it saturates to a flux balance (W1 cell: mean $u \rightarrow \approx 1.32$ by presentation ~ 30). The integration is over *the organism's own spiking*, which under drive is stimulus-locked but conflates patterns through the shared pool (§5).

Rung 5: intrinsic scalar u . What survives into silence is the integrated scalar field. Its information content is the subject of §7: it is a population rate trace, not a sensory memory.

6.1 The dilution result across repeated presentations

The loss that matters for persistent learning is not between rungs 1 and 2 but in what *repeated* presentations do to episode identity at the persistent rungs. The alternating-regime measurements [18] provide the sharpest quantitative statement: after each presentation, the state of the synaptic store after an A presentation and the state after the adjacent C presentation converge — the cosine of the per-neuron A-mass vector after C_k versus after A_k rises $0.972 \rightarrow 0.990$ over $k = 1 \rightarrow 40$. Simultaneously the alternating store becomes a *third configuration*, not a mixture of the trained endpoints (cosine to the A-only endpoint 0.277, to the C-only endpoint 0.235, while the endpoints are at 0.119 from each other; the best linear mixture leaves 96% of the alternating norm as residual) [18]. In other words: with alternation, episode identity is diluted to a superposition; with blocking, it is overwritten (§9); at the persistent rungs the store converges to quantities that cannot answer “which episode, most recently.”

The same convergence was first measured in u (alternating- u cosine $\rightarrow 1.0$) and then reproduced at the synaptic level — the records emphasize that the superposition failure is a property of the shared additive store, not of the particular substrate [18]. Figure 4 plots the two measured faces of the dilution: the converging adjacent-state cosine (left) and the within-block response-count decay (right), from the preserved alternating and blocked runs respectively.

Demonstrated: the persistent storage rungs (synaptic weights; integrated u) do not retain episode identity under repeated or alternated exposure; adjacent-episode state similarity rises toward 1 ($0.972 \rightarrow 0.990$).

Supported: the loss occurs at the transition from presentation-local (spikes, counts) to persistent (integrals, weights) representations; the shared-budget additive store is the mechanism (later confirmed by the CLLA-era results, §10).

Not established: any persistent rung that distinguishes recent episodes — this is the negative result that motivates the rest of the paper.

7 Intrinsic Persistence Is Not Sensory Memory

The slow depolarizing state u (V2.1) was the designated candidate for a persistent, intrinsic memory that does not depend on synaptic structure. Phase I’s evidence against that role is unusually complete because the audits measured what u actually does during and after the stimulus, and because one claim was retracted when its runs turned out not to exist.

7.1 What u is, measured

u tracks the organism’s own spiking through a first-order filter: per-spike increment β , decay τ_s . Three measured facts [22]:

1. **It saturates during the stimulus.** Means rise monotonically with cumulative exposure and plateau at a flux balance (W1: $\bar{u} \rightarrow 1.32$ by presentation ~ 30 ; W2: $\bar{u} \rightarrow 0.86$, still rising at drive end). A saturating integrator cannot encode cumulative episode history beyond its equilibrium.

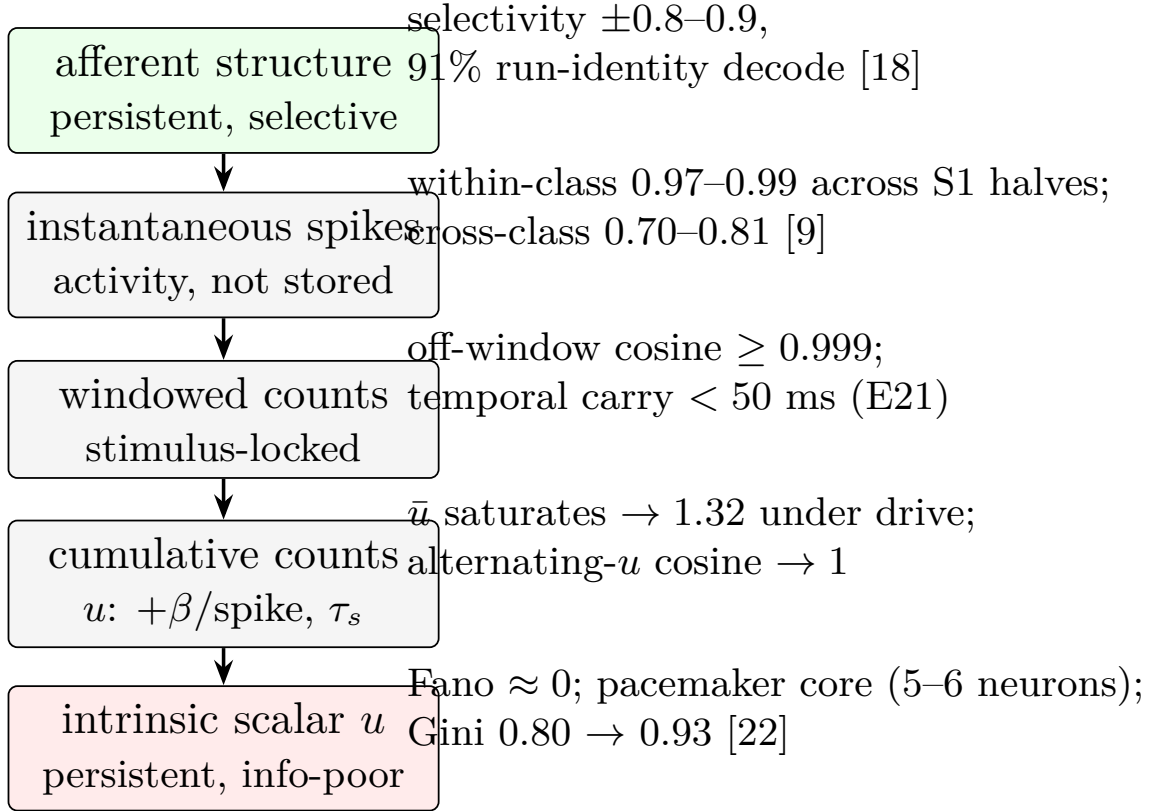


Figure 3: **The persistent-state information bottleneck.** The substrate’s automatic compression cascade: afferent structure (persistent, selective) $\rightarrow$ instantaneous spikes (activity) $\rightarrow$ windowed counts (stimulus-locked, not retained) $\rightarrow$ cumulative counts (the slow integrator u) $\rightarrow$ intrinsic scalar u (persistent but information-poor; §7). Each arrow is a lossy compression performed by the substrate’s own dynamics.

2. **Persistence without structure.** After stimulus removal the population does not relax: the active core self-sustains as deterministic pacemaking — 500 spikes/s on the refractory ceiling (2 ms refractory $\Rightarrow$ 500 Hz), Fano factors ≈ 0 , 5–6 active neurons of 52, Gini concentration 0.80 $\rightarrow$ 0.93 over silence. The persistent activity is a clock, not a code [22].
3. **The only tested claim of a discriminative u is retracted.** The V2.1 Stage-C validation table claimed temporal-capacity unchanged at the rule-selected operating point; the referenced runs do not exist in the repository. The information audit discovered the phantom reference, retracted the verdict as ungrounded (“unsupported, not refuted — unmeasured”), and added an integrity regression that fails any record citing nonexistent runs [22].

7.2 The historical wedge: u fails during the stimulus, not only after it

The standard memory objection to an integrator is that it decays after stimulus offset. The ANIMA record shows something stronger: in the regimes where episode identity was tested, the persistent quantities fail to distinguish episodes *while the stimulus is still present*. The alternating-store cosine after adjacent A and C presentations converges to 0.99 (§6) — the state after an A

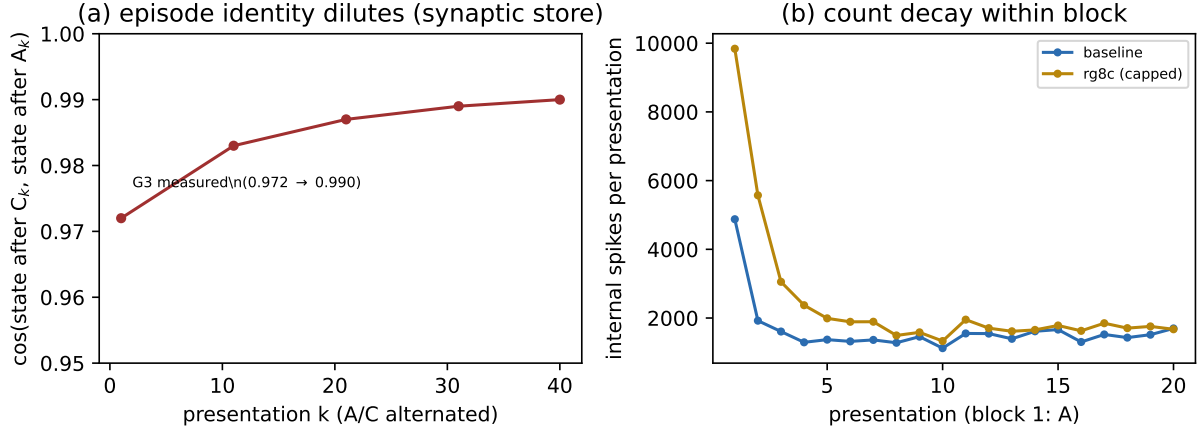


Figure 4: **Information dilution across repeated presentations.** Left: convergence of the synaptic store after adjacent A and C presentations under alternation (cosine $0.972 \rightarrow 0.990$; [18]). Right: per-presentation internal spike counts decay within a blocked block (baseline run), the count-level face of the same dilution. Both measured from preserved artifacts.

presentation is already near-identical to the state after the adjacent C , at a measurement taken 2 s apart with exactly one intervening presentation [18]. The V2.2 latch design (discussed next) was in part motivated by this timing: it attempted to make a per-neuron bistable bit the carrier, under u-driven SET/RESET thresholds — and failed as a carrier (latches decode the curriculum at chance in all 120 stage-2 runs; $\eta_{\text{rel}} = 1.0$ self-cancels) [19]. The E18–E21 temporal-capacity series bracketed the same wedge from the output side: no antecedent-dependent divergence over 800 ms; output responses stimulus-locked with zero endogenous output activity; temporal capacity below 50 ms [19].

Demonstrated: scalar intrinsic persistence (u) is self-sustaining but information-poor (saturated mean, pacemaker core, Fano ≈ 0); the V2.2 latch carrier failed 120-run tests; the temporal carry of the substrate is below 50 ms.

Supported: episode discriminability fails inside the stimulus window, hence the bottleneck is not post-offset decay but representation-integration collapse.

Not established / retracted: the Stage-C capacity claim is ungrounded and retracted; no A/C contrast in u was ever executed.

8 Synaptic Plasticity as Persistent Structure

If activity rungs cannot persist information, the weights must: they survive silence by construction (they are only changed by plasticity events), and the record shows they are the substrate’s persistent store. The plasticity-as-memory audit [18] established this with four independent verdicts.

Persistence (G1, G2). Cross-run training (A-only vs. C-only, same seed, identical initial weights) produces distinct persistent configurations: cosine 0.119, selectivity ± 0.8 –0.9, 91% leave-one-pair-out run identification. Selectivity develops monotonically through S1 and *continues to develop in silence* ($+0.903 \rightarrow +0.914$, driven by pruning of residual non-A mass — 617→309 live

afferents), while the matrix itself changes only $\sim 1.4\%$ across 20 s of silence. Structure persists; activity decays (activity rungs decay $3.5\text{--}7\times$ in the same window: u -norm $7.999 \rightarrow 2.282$, internal spikes $479/\text{s} \rightarrow 71/\text{s}$).

Create/maintain mechanics (G4). The causal records (E15–E17, SDE series) attribute organization to a specific subset of mechanisms: M2 normalization (per-neuron t_e budget) is *required* for the separated regime (M2-off destroys separation; E16), while STDP is not individually necessary for the afferent-selection trajectory (STDP-off reproduces it near-verbatim, E16), and M3/M4 topology dynamics are not necessary either (E17). Maintenance is M2 cadence plus M4 pruning; passive decay ($10^{-6}/\text{tick}$) is the dominant single eraser of first-block mass (SDE-C2: decay-off preserves the first cohort in 6/6 blocked runs; residual skew $\approx 0.07\text{--}0.09$ remains). Retrieval is a dynamics function: the REV1 phase-reversal experiment re-expresses an early-established configuration with zero plasticity events and $\leq 3\%$ weight change (E12–E17) — forward dynamics through stored weights.

What is not established at this layer (G8). Re-expression of a stored configuration by a later input exists at the association/trace level; *episode-specific* reinstatement under alternation does not — the alternating store superposes (§9) and no later A-presentation reinstates an A-configuration distinguishable from the adjacent C-configuration.

Demonstrated: synaptic structure is the persistent store; creation requires budget normalization; retention is bounded by decay and order; retrieval-by-forward-dynamics exists for trace-level configurations.

Supported: the information bottleneck is shared by every persistent substrate tested (weights and u show the same superposition signature), hence the limiting factor is the shared additive store under a shared budget, not the substrate identity [18].

9 Synaptic Interference: Isolated, Blocked, and Alternating Regimes

The regime trichotomy — isolated, blocked, alternating — is the organizing experimental fact of the synaptic-structure era. All three were measured on the same cell (the e24 operating point $\beta = 0.0046875$, $\tau_s = 5000$), same seeds, with the same snapshot-based instrument [18]; Figure 5 shows the end-state channel masses.

Isolated. A single pattern trained to completion yields clean winner selection: A-mass 0.290 / C-mass 0.026 (A-trained) or the mirror (0.031 / 0.239, C-trained). The unchallenged group is pruned away by budget normalization plus M4 selection.

Blocked. Sequential blocks overwrite: BAC (20 A then 20 C) ends at A-mass 0.068 / C-mass 0.137 — the first block’s mass is removed by the second (unmodified substrate; the CLLA-era numbers in §10 reproduce the same shape under protection). BCA ends at 0.234 / 0.105. Each blocked run drifts toward its final block’s trained endpoint (cos to its own endpoint 0.52–0.58) but lands between the endpoints (both blocks leave residual mass).

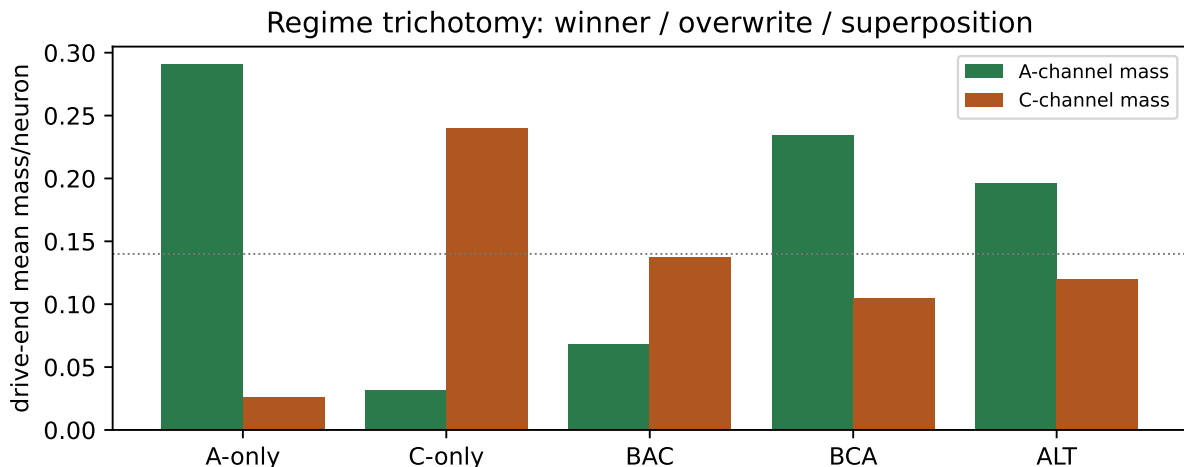


Figure 5: **Isolated vs. blocked vs. alternating synaptic organization** (same cell, same seed; drive-end mean per-neuron channel mass; single-pattern and blocked bars recomputed from preserved runs by the committed `papermass` instrument, reproducing the audit table values at print precision; the alternating bar is the e24-il run at the same 40-presentation convention (the audit’s longer 40+40 alternating run reads 0.239/0.162 — the same superposition, not a mixture of the endpoints) [18]). Isolated training selects one channel group; blocked order overwrites the earlier block; alternation retains both groups but as a superposition that does not discriminate the most recent episode [18].

Alternating. Both groups remain driven, so neither is pruned: end-state A-mass 0.239 / C-mass 0.162 — both retained, no overwrite. But the retained state is a superposition: it is neither a mixture of the endpoints (best linear mixture leaves a 96% residual) nor a discriminator of the most recent episode (adjacent-state cosine $0.972 \rightarrow 0.990$, §6).

Causal decomposition. SDE-B (doubling t_e) showed that capacity is not the lever: the surplus feeds the dominant trace and coexistence does not improve. SDE-C2 isolated passive decay as the dominant eraser of the first block (decay-off preserves first-cohort mass in 6/6 blocked runs). SDE-D showed the STDP a^-/a^+ asymmetry is secondary and context-dependent; the leading unresolved residual mechanism was M2 reallocation (inference at the time, later supported by the trace-partitioned M2 experiment, V2.3, which rescued blocked-order collapse-class cells 6/6 by capacity-matched per-bucket targets [19]).

Demonstrated: isolated $\rightarrow$ winner selection; blocked $\rightarrow$ recency overwrite; alternating $\rightarrow$ coexistence without episode discrimination. The trichotomy reproduces at the protected (CLLA) layer later (§10) and under all tested amendments except the M2 partition, which was tested blocked-only.

Supported: the shared per-neuron additive budget is the interference medium; passive decay is the dominant eraser; budget partition is the only measured lever improving blocked coexistence.

10 Consolidation-Locked Allocation (CLLA)

Consolidation-locked allocation (CLLA) is the Phase I attempt to make synaptic interference addressable: protect synapses that have reached permanence, so that later plasticity cannot silently dismantle an established structure. The sequence of four records is the clearest example of the program’s causal narrowing, and each step is reported separately because the amendments are protocol changes, not re-runs.

10.1 The mechanisms

Three mechanisms, separable in the record:

- **Consolidation.** At M3 permanence (candidate weight $\geq \theta_{\text{perm}} = 0.05$), a synapse may be flagged *consolidated* (`assembly_protect` on): it becomes exempt from M2 rescaling, M4 silence-pruning, and passive decay.
- **Bounded protected mass.** Consolidated LTP is clipped so that per-neuron protected mass $P_i = \sum_{\text{consolidated}} w$ satisfies $P_i \leq p_{\text{max}} t_e = 0.60$ at every tick (the bounded-protection amendment; [4]).
- **Residual allocation gate (later).** Candidate accumulation gated by the protected-current fraction $R_i = I_{p,i}/(I_{p,i} + I_{w,i})$, so unconsolidated (“unexplained”) drive keeps allocation priority; §11.

10.2 The four-record sequence

1. Unbounded protection — runaway. With consolidation but no growth bound, protected synapses are exempt from every brake (M2, decay) while STDP still applies: within-pattern LTP runs unimpeded, P reaches $\approx 7.9/\text{neuron}$ against the 0.60 cap (cap violations 2,284–3,408 per surviving run; $\max P/(0.75t_e) = 13.3$), and 6 of 12 runs abort on the runaway detector (50 Hz/5 s). Verdict: bundle NOT SUPPORTED, failure mode F3/F4 — protection without a homeostatic bound is unstable [6].

2. Bounded protection — viability. The amendment clips consolidated LTP at the remaining headroom. All 12 runs complete; protected mass pins at 0.6000 (f32-exact) everywhere; working budget stays strictly positive ($t_e - P \geq 0.2$); rates settle to 3.5–6.5 Hz; the previously-aborting cells complete. Verdict: dynamically viable [4].

3. Capability — coexistence split. The frozen falsifier F1 requires drive-end channel mass $\geq 0.5 \times$ per-seed single-pattern reference in all arms. Alternating (il) passes 3/3 (protected-ratio A/(A+C) 0.53–0.59); blocked (bac/bca) fails 6/6: the later block captures the shared working budget and the protected slots (bac ends C-poor: C 0.101/0.057/0.037; bca ends A-poor: A 0.116/0.056/0.055). F3 (cap), F3b (exhaustion), F4 (stability), F5 (no hidden context), F7 (protected-mass retention, ratios 0.80–1.00) pass 12/12. F2/F6 are unmeasurable under the frozen windows (empty by curriculum construction — a recorded protocol defect, §4); the closest-available-window diagnostics are consistent with late separation (cross 0.15–0.40, within 0.85–0.89 in all three il runs). Verdict: bundle NOT SUPPORTED, with the failure localized to *allocation* — blocked-order recency capture — not protection, stability, or retention [5].

4. Interpretation split. The records draw the distinction that organizes the rest of Phase I:

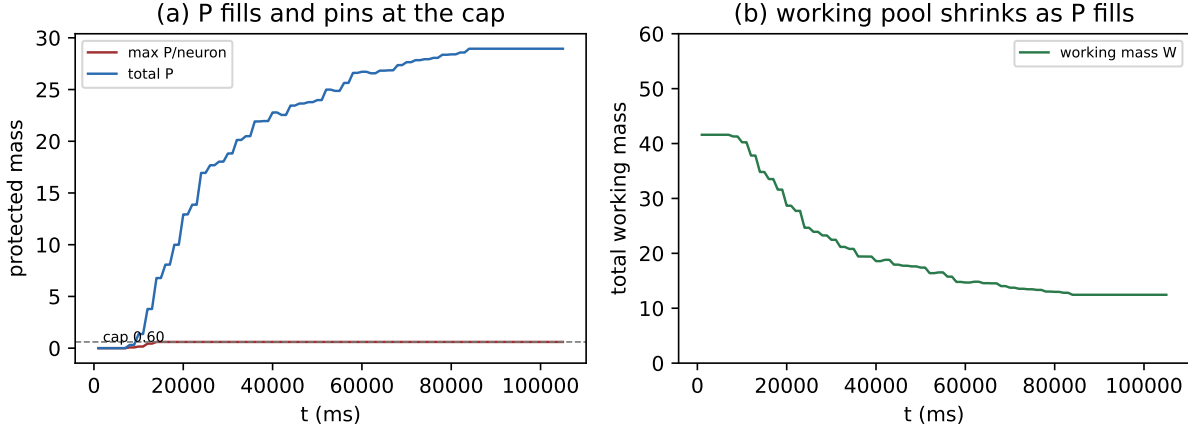


Figure 6: **CLLA bounded protection: stability at the cap.** Per-snapshot (1 s) trajectories of the previously-aborting s424242-il cell under the bounded amendment: per-neuron protected mass P rises to and pins at the 0.60 cap; total protected mass grows as more neurons fill; the working pool W shrinks to the residual budget as protection fills; zero failures (all 12 runs complete) [4].

Protection (bounded) is established: retained, capped, stable.

Coexistence is order-dependent: demonstrated under alternation, absent under blocked order.

Allocation — who gets the budget and the protected slots when patterns arrive sequentially — is the unresolved capability that the remaining experiments attack.

Figure 6 shows the bounded-protection witness trajectory: protected mass fills to the cap and pins there while the working pool shrinks, without instability.

11 Allocation Experiments

With stability established, the frozen question became allocation: *can a local mechanism prevent first-arrival starvation of a later distinct configuration?* Each registered experiment tightened the answer and moved the identified bottleneck one step downstream. All five share the identical identity gate (flag-off byte-identity 9647ea8a0ca4dbd2), the identical 12-run matrix (bac/bca/il/d $\times$ three seeds), and the identical primary endpoint: second-block protected mass $\geq 0.5 \times$ first-block protected mass.

11.1 Residual allocation gate (rule)

The gate (protocol [1] extension, record [3]) multiplies candidate accumulation by $g_i = \max(0, 1 - R_i)$ with headroom > 0 , where $R_i = I_{p,i}/(I_{p,i} + I_{w,i})$ is the protected-share of the neuron’s delivered input current. The implementation behaves exactly as specified — the gate sat at $g = 1.0$ through the entire second block (R -median 0.000), headroom was reserved (0.249–0.294 vs. 0.155–0.161 in the no-rule runs), and first-block permanence decelerated (170 events, 8.5/pres, vs. 364, 18.2/pres) — yet the primary endpoint failed 6/6 (ratios 0.008–0.192 vs. the 0.5 bar). The trajectory instrument located the failure *upstream of the rule*: the second-block working substrate is destroyed during the first block (live unconsolidated C-channel afferents $197 \rightarrow 77$ under M2

normalization of the shrinking working budget and M4 pruning), so at second-block onset $g = 1$ with almost nothing to accumulate. Verdict: the gate is not the bottleneck; substrate availability is [3]. Alternating coexistence preserved (F1-il 3/3).

11.2 Dormant-candidate reserve

The reserve (record [8]) lets ever-coactive candidates survive dormancy at a floor instead of dying, protecting pre-associations across inactive periods. The mechanism works measurably — 90–255 reserved candidates per run survive — but at second-block onset in the blocked arms the reserved C-cohort (channels 8–15) is exactly *zero* (mean over the A block: reserved A-cohort 123.7, C-cohort 0.0); the block-2 permanence count is 38 vs. 35 without the reserve. The record classifies the outcome as *neither* preservation failure *nor* flux failure: a trigger keyed on “ever-coactive” is satisfiable only by the pattern that has already been active. The reserve is vacuous for a genuinely first-seen pattern by the semantics of its own trigger [8].

11.3 First-exposure allocation

The next amendment biases candidate redraw/eviction toward channels that fired in the current window, so a currently active novel input can bind candidate slots before churn eliminates its substrate (record [12]). Primary endpoint still failed 6/6 (ratios 0.072–0.216), but the causal path became exact: first C permanence at $t = 47,400 - 2.4$ s after block onset, $10\times$ faster than baseline; permanence events 41 vs. 35 (allocator-only) and 38 (reserve); block-1 allocation *increased* (248 vs. 170 events — the mechanism strengthens whatever is currently active). The binding supply was capped by two interacting facts: only ≈ 0.71 visible channels/neuron survived at onset (visibility cap), and the redraw-bias branch was structurally inert (`connected()` excludes exactly the visible channels it would have preferred). The implementation audit ([13]) identified the latter as a spec vacuity — the bias could never fire — and the next amendment corrected the source.

11.4 Source-selection correction

The corrected source reads the module’s per-window firing record instead of afferent-derived channel sets, making all firing channels observable regardless of synapse survival (record [15]). Supply tripled ($41 \rightarrow 61$ C-cohort permanence events); first permanence at 2.5 s; primary endpoint still failed 6/6 (ratios 0.046–0.193). The classification that survived into the paper’s central question: flux improved but remained asymmetric (3.05/pres C vs. 13.2/pres A), and the measured residual bottleneck was *post-permanence weight growth + working-afferent churn* — newly protected second-block synapses consolidate at $w \approx 0.02$ and grow little (mean 0.054 vs. the first block’s 0.096) because the working C afferents that would drive their LTP keep being destroyed ($29 \rightarrow 11$ across the second block itself) by M2’s working-mass squeeze. The record registered the two candidate amendments it did not run (e1: consolidation-time protection of a working-afferent fraction; e2: protected-path LTP growth) [15]. Figure 7 plots the final state of this decomposition.

11.5 What the sequence ruled out

- Allocation gating: **ruled out as the limiter** — the gate did nothing wrong and changed nothing.

- Dormant preservation: **ruled out** — preservation works, but has nothing preservable before first contact.
- First-exposure visibility: **fixed then ruled out** — the source correction removed the vacuity; supply tripled.
- Flux (candidate accumulation rate): **supported as secondary** — asymmetric (3.05 vs. 13.2/pres) but not the binding constraint.
- Working-substrate churn against the *new* protected structure: **the measured residual** — leads directly to the recruitment question (§12).

12 The Novel-Pattern Recruitment Bottleneck

The allocation sequence ended with a measured paradox: the corrected allocator could deliver candidate substrate (permanence supply tripled), and yet the second block’s protected structure stayed at $\approx 1/10$ of the first block’s mass. The resolution required decomposing what a newly consolidated synapse needs *after* permanence: LTP support from co-firing postsynaptic activity. Two read-only audits [20, 14], both over the committed source-correction runs (registration-era baselines of §13), localized the failure to the very first link of the second block.

12.1 Event-level decomposition (weight-growth audit)

For the 61 C-cohort permanence events of the second block in the baseline blocked run, the per-synapse decomposition is complete [20]:

- starting weight at permanence: $w_{c,perm} = 0.02$ (snapshot-verified);
- pre-spikes: abundant — C channels fire 56–87 spikes per 500 ms presentation; the input side is not starved;
- post-spikes: the decisive deficit — the last A presentation fires 1,697 internal spikes; the first C presentation fires 101; C presentations sustain 101–222 (a $17\times$ collapse at the block switch; A-block presentations run 1,300–1,700);
- applied LTP: the STDP balance on the C cohort over the C block is *net negative* (-21.63 over 1,961 coalesced events): with post activity so sparse, the dominant pre/post pairing is LTD;
- per-synapse growth is intact whenever posts exist (measured factors of $2.5\text{--}3.8\times$, e.g. $0.02 \rightarrow 0.185$);
- M2 is not suppressing: the working target stays $0.360 \rightarrow 0.303$ across the block (per-neuron P $0.44 \rightarrow 0.50$, cap 0.60);
- M4 is quiet in the second block: 23 C-afferent prunes vs. 161 in the first block;
- causal order: the post-activity collapse (pres-21) precedes every C permanence (first at $t = 47,500$, 2.5 s after onset) and all C weight growth.

Verdict: **event-limited (A)** — second-block C fails at the level of relevant postsynaptic LTP opportunities; the growth machinery works where posts exist.

12.2 Drive decomposition and the IL control (drive audit)

Why do C presentations produce a hundred spikes instead of a thousand? The drive audit [14] decomposed the first-exposure currents (per-neuron means, snapshot + spike telemetry):

quantity (first exposure, per neuron)	first-A (block 1, pres 1)	first-C (block 2, pres 21)
afferent current I_{aff}	11.98	1.57
recurrent current I_{rec}	6.90	0.05
inhibitory current I_{inh}	-2.19	-0.81
posts/n	0.98	0.12

The first-C presentation delivers $\approx 7.7\times$ less afferent current than the first-A presentation did, all of it working (unconsolidated): block-1 churn (M2 working-budget squeeze + M4 pruning, dominated by passive decay per SDE-C2) reduced the live C working afferents from ≈ 197 to ≈ 29 before C was ever presented. With 1.6 drive/neuron and no recurrent support, the population sits below threshold; no posts; no recurrent bootstrap; the LTD-dominant STDP balance; protected structure starves. M6 inhibition is excluded as the cause: it is $2.7\times$ *lower* at first C than at first A (it follows activity, it does not create the deficit).

The control that converts this from correlation to causation is the interleaved regime. In the committed alternating runs, C is present from presentation 1, so its substrate is never churned; first-C afferent current is 11.13/neuron, recurrent 7.05, posts 1.00 — numerically identical to A’s first contact — and C permanence (98) matches A (95) [14]. The blocked-vs-alternating comparison isolates *prior representation* / *substrate competition* as the variable: the C stimulus itself is intrinsically fine.

Demonstrated: first-exposure drive for a late pattern is $\approx 8\times$ below first-block levels because its working substrate was destroyed before exposure (prunes +18–31%; passive decay family per SDE-C2); posts collapse $17\times$; the mechanism is event-limited, not LTP-, M2-, or M4-limited.

Supported: the deficit is curriculum-inherent (blocked-order churn); the alternating regime recruits symmetrically without any added mechanism.

13 The Fixed- k_g Recruitment-Gain Experiment

The recruitment audits localized the bottleneck; the fixed-mechanism experiment tested the smallest hypothesized amendment: a local, novelty-dependent recruitment gain that transiently raises a neuron’s recruitability when its input is unexplained and its post activity is sparse. The design review [17] derived the form from existing local quantities only — the allocation rule’s unexplained fraction $1 - R$ and the neuron’s own working input current I_W — and concluded that a single new dimensionless magnitude k_g was unavoidable (every existing constant is $\leq 1\times$; the required bootstrap is $\geq 6\text{--}10\times$ under the measured drive→ posts map, whose middle band [4.2, 19] current units is unmeasured by any committed presentation). Per the pre-registration mandate the parameter was not fitted: the user froze the first mechanism test at the lowest defensible value,

$$i_{\text{boost},i}(t) = k_g \cdot \max(0, 1 - R_i(t)) \cdot I_{W,i}(t), \quad k_g = 8.0, \quad (4)$$

with everything else byte-identical to the committed configuration, including the identity gate and the frozen endpoint bars [17]. The experiment was executed in two registered arms.

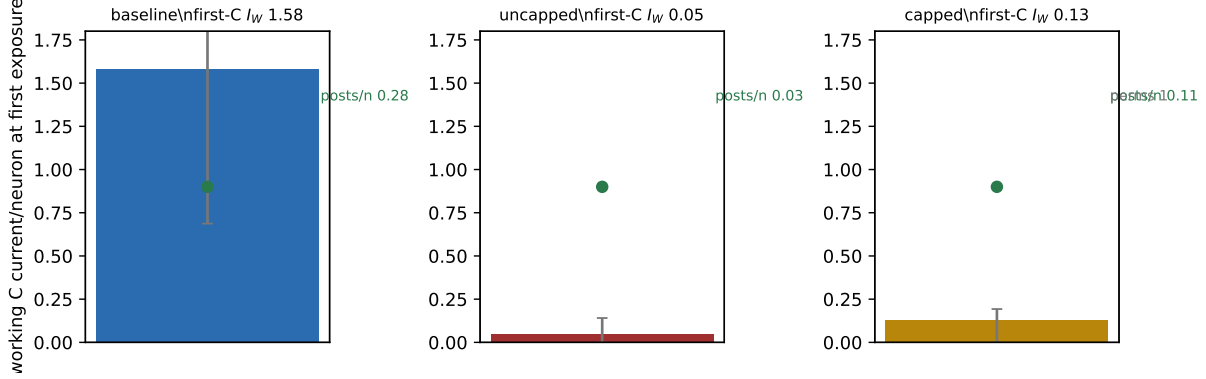


Figure 7: **Allocation/recruitment failure decomposition** at first exposure of the late pattern (blocked runs, three seeds; bars show mean over seeds, ticks the range). Left: working afferent current surviving to first exposure (baseline 1.565/neuron) collapses under the uncapped gain and is only partially rescued by the cap. Center: first-exposure posts/n. Right: second-block permanence events (bar 150). All panels from preserved runs via the committed `rg8eval` instrument [16, 21].

13.1 Registration A: uncapped ($k_g = 8.0$)

Nine gain arms (bac/bca/il $\times$ three seeds) against the nine committed baselines, all flag-off byte-identical [16]. Endpoints: 4a zero runaway trips (PASS, failures [] in all arms); 4b block-1 first-exposure burst band (FAIL — pres 1–5 mean 185.9–201.0 Hz vs. the [100, 200] Hz bar, with pres-1 at 391–449 Hz vs. 57–94 baseline); 4c first-pattern posts at presentation 20 preserved (PASS, ratio 1.00–1.04); 4d protected cap (PASS, pinned at 0.6000 as in baseline); and the three primary recruitment endpoints (all FAIL): first-novel posts/n ≥ 0.5 (measured 0.00–0.19), novel permanence ≥ 150 (0–22), protected novel end-mass ≥ 0.09 (0.000–0.058).

The mechanism of failure was measured, not inferred: the full-gate boost on the *first block's own* first exposures ($I_W \approx 12$, $R = 0 \Rightarrow$ boost ≈ 96 , drive ≈ 108 vs. the normal 38) drove pres-1 bursts to 391–449 Hz; the hyperactive block then *accelerated* the churn that destroys the late pattern's substrate (C-cohort prunes in block 1: 195–215 vs. 164–187), so at first C exposure the surviving working current was 0.000–0.14/neuron vs. 1.565 baseline — and the boost is input-proportional: $I_W = 0 \Rightarrow$ boost = 0 by (4) itself. The mechanism never engaged at its target moment. Per the frozen interpretation rule, a 4b failure classifies as a *recruitment-gain operating-point failure first*; the weak-pattern recruitment hypothesis was not falsified by registration A [16].

13.2 Registration B: threshold-completion cap

The separately registered amendment [21] clamps the boost at the current needed to reach threshold in one step,

$$i_{\text{boost},i}(t) = \min\left(k_g \cdot \max(0, 1 - R_i(t)) \cdot I_{W,i}(t), \max(0, (v_{\text{th}} - v_i(t)) \tau_m/dt)\right), \quad (5)$$

parameter-free (existing $v_{\text{th}} = 1$, $\tau_m = 20$, $dt = 1$); the clamp was implemented per-*neuron* after a per-synapse first pass was caught in review, with a discriminating unit test (two afferents, summed raw boost 41.6 > cap 20 $\Rightarrow$ one clamped boost). Identity gate and suites passed; all nine arms completed, zero failures.

Results against the same bars [21]: 4a PASS; 4b aggregate 5-presentation means 175.6–189.1 Hz inside the band — but pres-1 remained 378–440 Hz, and under the strict per-presentation reading 4b is still outside the band in every arm; the cap moved pres-1 only 391 $\rightarrow$ 378 Hz. The mechanism is now understood: the per-tick cap tops a below-threshold neuron to threshold on every tick while input is ongoing, sustaining ≈ 1 spike/tick (refractory-limited) regardless of boost magnitude — the overshoot is rate-sustained, not magnitude-sustained. The primary endpoints still failed everywhere (first-novel posts 0.00–0.19; novel permanence 5–15; protected novel end-mass 0.000–0.092, reaching 0.09 in exactly one arm with only 5 protected synapses); IL preservation stayed partial (C posts 1.08–1.21 $\times$, permanence 97/74/105 vs. the $\pm 20\%$ of 98 band); gap activity remained elevated (2.3–27.0 vs. 0.45–5.94 Hz baseline).

13.3 Why the experiment was closed

Per the pre-registered outcome rules, each arm was closed at its failure: no reparameterization, no bracketing, no post-hoc threshold change, no E-number. The honest verdict language of the records is preserved here:

Registration A: **operating-point failure** — 4b fails; the weak-pattern recruitment hypothesis is *not falsified* (the mechanism never fired at first exposure because its own block-1 side effect destroyed the substrate it amplifies).

Registration B: the cap repairs the aggregate operating point but not the first-exposure rate; the primary endpoints fail everywhere. The measured reason is structural: churn precedes exposure (even capped, block-1 prunes of the late cohort run 193–221 vs. 157–187 baseline), and the boost is input-proportional, so a membrane-side gain has almost nothing to amplify at first contact (surviving working current 0.00–0.19/neuron vs. 1.565). Under the registration B outcome rule (4b pass + primary fail $\Rightarrow$ falsified at this substrate), the records state the hypothesis as *falsified at this substrate under the corrected operating point*, while noting that neither registration achieved a clean block-1 operating point, so the falsification claim is bounded by the residual operating distortion [21].

Both registrations were closed per protocol; the threshold-completion cap is recorded as the designated, tested alternative; no further parameter was selected from the results.

14 Phase I Synthesis

Phase I’s net result is a capability map with a single open frontier. Table 2 states the capability, the status, and the record that establishes or withholds it. Each entry was verified against the cited record before inclusion.

The causal narrowing itself is the paper’s thesis, and Figure 8 presents it as a chain of measured limits:

1. the substrate organizes (afferent selectivity ± 0.9 , 91% run-identity decode);
2. transient representations exist (stimulus-locked counts; separation resisted E3/E3b dynamics interventions);
3. long-horizon accumulation destroys episode identity (adjacent- state cosine $\rightarrow 0.99$; LOPO fails under alternation);

Table 2: **Phase I capability matrix.** Statuses: *established* (measured, reproduced), *supported* (consistent evidence, non-decisive protocol slots), *insufficient* (measured, failed its role), *demonstrated under condition* (order-dependent), *not established*.

Capability	Status	Record
Sensory organization (afferent)	established	G1/G2 [18]
Transient representation (per-presentation counts)	established	E1 [9]
Synaptic reorganization	established	G1/G4 [18]
Scalar intrinsic memory (<i>u</i>)	insufficient	info audit [22]
Temporal (sub-50 ms) episode carry	not established (measured < 50 ms)	E21 [19]
Protected synaptic structure	established	bounded CLLA [4]
Alternating coexistence	demonstrated	CLLA F1-il [5]
Blocked-order coexistence	not established (overwrites)	F1-blocked, G3 [5, 18]
Sequential allocation	unresolved	rule/reserve/fe/fec [3, 8, 12, 15]
First-exposure substrate survival	not established	fec residual (e) [15]
Novel-pattern recruitment	unresolved	rg8/rg8c [16, 21]
General retrieval (episode-specific)	not established	G8 [18]
Closed-loop memory behavior	not established	E19–E23 [19]

4. scalar intrinsic persistence is information-poor (saturated integrator, pacemaker core, Fano ≈ 0 ; latch carrier fails);
5. synaptic structure becomes the persistent substrate (structure persists, activity decays; decay is the dominant eraser);
6. synaptic interference appears as order dependence (isolated selects, blocked overwrites, alternating superposes);
7. protection restores stability (bounded CLLA: cap pinned, zero failures) but not allocation;
8. allocation becomes limiting (gate, reserve, first-exposure: each moved the identified site downstream);
9. first-exposure allocation becomes limiting (source correction tripled supply; post-permanence growth starved);
10. novel-pattern recruitment becomes limiting ($17\times$ post collapse; $8\times$ drive deficit at first contact; IL control symmetric);
11. Phase I closes with that capability unresolved (both gain registrations closed by pre-registered rules; hypothesis bounded, not destroyed).

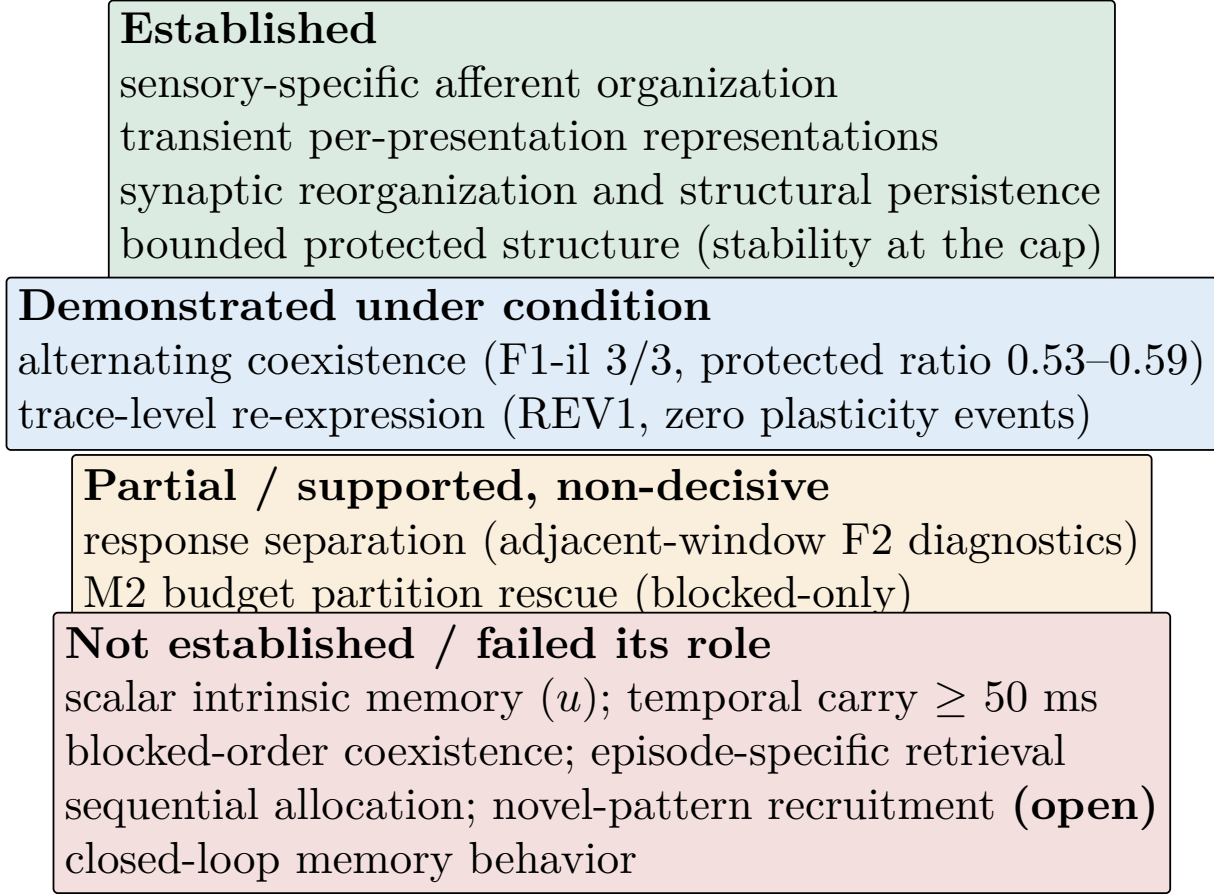


Figure 8: **Phase I capability map.** Green: established. Blue: demonstrated under the alternating condition. Amber: partial / supported-but-non-decisive. Red: not established / failed its assigned role. The chain reads as the causal narrowing of the persistent-learning failure described in the text; the final cell (novel-pattern recruitment) is the open frontier.

Each link is a registered experiment or read-only audit with its run record; no link is asserted from correlation alone (the IL control and the pre-exposure-vs.-during-exposure ordering supply the causal anchors).

15 Limitations

The record is deliberately narrow, and the following limitations bound every claim in this paper.

Scale and stimulus regime. All CLLA-era results use one organism size (52 internal/output neurons), three seeds, 8-channel synthetic Poisson patterns at 20 Hz, and one operating cell ($\beta = 0.0046875$, $\tau_s = 5000$ ms). Scale studies (E7/E8) exist but were not part of the allocation and recruitment sequence; no claim about larger substrates is made.

No closed loop. Nothing in Phase I closes a behavior loop. E19–E23 showed output activity is stimulus-locked and (in E23) that consequence shaping exists but inverts; closed-loop memory

behavior is listed as not established.

Protocol defects bound several verdicts. The CLLA capability F2/F6 verdicts are formally unmeasured (empty frozen windows; adjacent-window diagnostics only). The V2.1 Stage-C claim is retracted. The fe-era afferent-derived source was vacuous and was corrected; the recruitment design review’s drive→posts map has an unmeasured middle band, so bootstrap magnitudes were bracketed, not calibrated. Each is reported where it matters.

Operating-point confounds. First-exposure burst overshoots (baseline D arm peaks 138.8 Hz; gain registration pres-1 378–449 Hz) sit near the runaway threshold. The gain experiments were closed with the operating point not fully clean under the strict 4b reading, which bounds the strength of the falsification claim recorded for registration B.

Theoretical claims are minimal. The paper does not claim biological fidelity, optimality, or any statement about what real nervous systems must do; the substrate is four LIF populations with textbook plasticity, studied as an experimental system. Interpretive claims (“the shared additive budget is the interference medium”) are labeled supported-versus-hypothesis in the text.

What is not claimed about failure. “Not supported” and “not established” are never converted into “impossible.” In particular: the recruitment-gain hypothesis is bounded by two registrations at one magnitude; other operating points, other gain forms, and substrate-preserving mechanisms are untested. Alternating coexistence being demonstrated does not establish general memory capability; it establishes coexistence under alternation.

16 Discussion

16.1 Representation

Phase I demonstrates that a minimally specified spiking substrate develops sensory-specific organization (afferent selectivity ± 0.8 – 0.9 ; 91% run-identity decode of per-neuron afferent vectors; repeatable per-presentation count vectors). It equally demonstrates that the transient representation is weak where it matters: cross-pattern response separation stalled near 0.67–0.74 against a 0.60 bar, and the two dynamics interventions (adaptation, lateral inhibition) moved it in the wrong direction [9, 11]. The binding problem of this organism is representational: 24 afferents drive one 40-neuron recurrent pool through random sparse wiring, so every readout mixes patterns. Nothing in the record suggests this is fixable by further rate dynamics; the only lever that has ever moved coexistence is budget partition (V2.3, blocked-only) [19, 18].

16.2 Persistence

Persistence has two measured faces. Structural persistence is real and bounded by mechanics: weights survive silence (matrix drift $\sim 1.4\%/20$ s), decay is the dominant eraser (SDE-C2), and bounded protection stops the runaway that unbounded protection caused [4, 6]. Activity persistence is information-poor: the V2.1 slow state self-sustains but saturates, and its endogenous form is deterministic pacemaking [22]. The paper’s strongest persistence claim is negative and precisely delimited: no persistent substrate measured in Phase I retains episode identity through repeated or alternated exposure — including, in the testable case, during the stimulus itself.

16.3 Interference

Interference is order-dependent and budget-mediated. Isolated training selects; blocked training overwrites; alternation superposes into a third configuration. The causal probes attribute the erasure to passive decay acting through M2’s shrinking working budget, with M4 as the executing pruner; STDP’s a^-/a^+ asymmetry is secondary (SDE-D). Protection fixes stability, not interference: even with bounded protected structure, blocked order captures the budget for the latest block [5]. Interference is therefore best described as an *allocation* phenomenon — competition for a shared local budget — rather than a plasticity-sign phenomenon.

16.4 Allocation

The allocation sequence is the paper’s procedural core: five registered experiments, each gated byte-identical, each moving the identified bottleneck one step downstream (gate $\rightarrow$ trigger semantics $\rightarrow$ visibility $\rightarrow$ source $\rightarrow$ post-permanence growth). The final measured residual is that the working substrate of a not-yet-presented pattern is dismantled before its first exposure, and that even a corrected allocator cannot grow the protected structure once substrate is gone. This is the point where Phase I’s causal narrowing arrives at the architectural frontier: allocation works when the pattern is active, and fails for patterns that have never been active — which is exactly the class that sequential memory requires.

16.5 Recruitment

The recruitment deficit is the frontier’s content: a genuinely new late-arriving pattern cannot recruit postsynaptic activity because its afferent substrate was churned away, and no membrane-side, input-proportional gain can amplify a substrate that is not there. The IL control proves the stimulus is not at fault. Two registered mechanism tests bounded the hypothesis without resolving it; the threshold-completion cap fixed the aggregate operating point but could not repair first-exposure rate or substrate survival. The discussion stops here by intent: Phase I’s evidence supports a direction (architectures for multiple independently retrievable persistent representations under sequential experience), not a design.

17 Conclusion

ANIMA Phase I investigated emergent organization and the architectural limits of persistent learning in a minimally specified spiking nervous system. Under frozen protocols, byte-identical identity gates, and preserved negative results, the record demonstrates that:

- sensory-specific synaptic organization emerges from local plasticity without supervision;
- transient spike-count representations exist but separate poorly across patterns;
- every persistent substrate tested — the slow intrinsic state and the synaptic weight structure alike — fails to retain episode identity through repeated or alternated exposure, in the testable case already during the stimulus;
- bounded consolidation produces stable protected structure and preserves alternating coexistence, but blocked-order allocation remains recency-dominated;

- the allocation sequence localized the residual failure to the working-substrate churn that precedes a never-seen pattern’s first exposure, and from there to the recruitment of postsynaptic activity for genuinely novel late-arriving patterns.

Phase I closes with that capability unresolved. The final experiment tested the smallest hypothesized local amendment — a fixed-magnitude, input-proportional recruitment gain — and was closed by its pre-registered rules at two registrations, having established the operating-point failure mode and the structural bound (a gain cannot amplify a substrate destroyed before exposure), without either destroying or confirming the weak-pattern recruitment hypothesis for the class of mechanisms. What remains is not a tuning problem within the tested architecture but an architectural question: how a substrate whose persistent store is a shared additive budget can form multiple independently retrievable persistent representations under sequential experience.

18 Future Work

Phase II design is deliberately out of scope. What Phase I supports is a single directional statement:

The open capability is *multiple independently retrievable persistent representations under sequential experience*. Future work should investigate architectures that make sequential, late-arriving patterns acquire (i) a surviving working substrate at first exposure, (ii) postsynaptic recruitment adequate to establish protected structure, and (iii) episode-distinguishable retrieval — in that dependency order, which is exactly the narrowing Phase I measured.

Three recorded, unexecuted candidate directions belong to this statement and are listed here only as evidence-adjacent hypotheses, not as designs: (a) protecting a never-yet-exposed cohort’s working afferents from M2/M4 churn during other blocks (the e1-family question registered in the source-correction record [15]; the two gain registrations supplied the direct evidence that substrate pre-exposure survival is the binding constraint); (b) a recruitment mechanism that is not input-proportional at the membrane (the measured bound $i_{\text{boost}} \leq k_g I_W$ is the reason both registrations could not engage); and (c) budget partition re-keyed along the channel-group axis, tested under alternation — the only mechanism ever measured to improve coexistence at the synaptic level, currently tested blocked-only [18]. Any of these, pursued, requires the full discipline: frozen protocol, identity gate, preserved failures.

19 Reproducibility and Artifact Availability

19.1 Repository structure

The experimental record lives in the project repository: `configs/` (frozen configurations, one file per arm), `crates/` (substrate and instruments; read-only analysis examples including `cross_cosine`, `read_rates`, `rg8eval`, `c1la_traj`, and this paper’s `papermass`), `docs/` (protocols, execution records, audits, correction records; filename and status conventions in §A), and `runs/` (preserved runs `runs/<exp_id>-<UTC>Z/` with `telemetry/` (deterministic event chunks), `snapshots`. `bin.zst` (1 s state), `metrics.json`, `report.md`; `runs/` is disk-only and gitignored, and an integrity regression fails any record that cites a nonexistent run [19]).

19.2 Lineage and commit discipline

The record is linear and amendable only by user-approved protocol changes. The most consequential corrections, with their provenance:

- **V2.1 Stage-C retraction.** A validation table citing nonexistent runs was committed; the information audit found the phantom reference, retracted the verdict as ungrounded (unsupported, not refuted), and added the phantom-reference integrity regression [22].
- **CLLA F2/F6 frozen-window emptiness.** The frozen windows (presentations 21–40 of each pattern) are empty under the frozen interleaved curriculum (20 presentations per pattern); the capability record reports the falsifiers as unmeasurable under the frozen definition and publishes adjacent-window diagnostics as non-decisive [5].
- **First-exposure allocator source vacuity.** The redraw bias could never fire as specified (visible `connected()` channels are exactly those the bias would select); the implementation audit named the vacuity, and a user-approved source correction (replace afferent-derived channel sets with the module firing record) tripled the measured bind supply ($41 \rightarrow 61$ permanence events) [13, 15].
- **Drive-gate dimensionality (X-series).** The I_{aff} gate used bare weight sums instead of membrane-scale current (`amplitude` $\times$ w); the mechanism review corrected the definition (reusing the existing v_{th} , no new parameter) before the CLLA-era drive analyses [19].
- **Recruitment-gain cap implementation.** Registration B’s cap was first implemented per synapse; review caught that a per-synapse clamp cannot bind a per-neuron sum; the two-pass per-neuron clamp was implemented and discriminated by a unit test before the matrix ran [21].

19.3 Reproducible analysis

All quantitative claims in this paper trace to preserved runs and committed instruments. The figure scripts in `paper/scripts/` rerun the instruments on the preserved runs and draw the figures; Table 3 collects the key quantitative anchors with their sources.

19.4 Determinism

Same seed $\Rightarrow$ byte-identical telemetry (single-run determinism is a substrate guarantee, verified by duplicate-run hash equality in the E1 era and by the flag-off identity gates of every CLLA-era implementation [19, 6]).

A Experimental Lineage

The experimental eras and their verdict records, in commit order. Era boundaries follow the reasoning eras, not the E-numbers.

Era	Experiments	Verdict	Record
Substrate + representation	E1–E11	organization forms; binding problem	[9, 11, 19]
Path dependence	E12–E17	structure is passive; M2 required	[18]
Capability failures	E18–E21	no temporal state; closed loop absent	[19]
Slow state	V2.1 (A,B; C retracted)	persistence without information	[22]
Latch	V2.2	carrier fails (120 runs)	[19]
Regime capacity	E24	NOT SUPPORTED (all contrasts $p = 1.000$)	[10]
Erase mechanism	SDE-B/C2/D	decay dominant eraser; M2 residual	[19]
M2 partition	V2.3	partial support; rescues blocked collapse (bca 6/6)	[19]
CLLA (unbounded)	CLLA bundle	NOT SUPPORTED (run-away; F3/F4)	[6]
CLLA (bounded)	amendment	dynamically viable (V1–V4)	[4]
CLLA capability	12-arm matrix	NOT SUPPORTED (F1 blocked; F2/F6 protocol defect)	[5]
Allocation rule	rule 12-arm	NOT SUPPORTED (substrate destroyed upstream)	[3]
Dormant reserve	reserve 12-arm	NOT SUPPORTED (trigger vacuous for novel)	[8]
First exposure	fe 12-arm	NOT SUPPORTED (visibility + inert redraw)	[12]
Source correction	fec 12-arm	NOT SUPPORTED (residual: post-permanence growth + churn)	[15]
Weight growth	audit (read-only)	event-limited; 17× post collapse	[20]
Drive audit	audit (read-only)	8× drive deficit; IL control symmetric	[14]
Recruitment gain A	fixed $k_g = 8$	operating-point failure; hypothesis not falsified	[16]
Recruitment gain B	+ threshold cap	primary endpoints fail; hypothesis bounded	[21]

Correction lineage (scientifically relevant): Stage-C retraction (ungrounded verdict) → phantom-reference regression; F2/F6 frozen windows unexecutable → adjacent diagnostics, non-decisive; allocator redraw vacuity → source correction (supply 41 → 61); drive-gate dimensionality → $I_{\text{aff}} = \Sigma(\text{amplitude} \cdot w)/v_{\text{th}}$; rg8c cap per-synapse → per-neuron two-pass. Each is preserved in the record documents, never silently repaired.

B Notation

The notation below is used consistently throughout the paper and the records. Implementations may differ in scale conventions (e.g. network-total vs. per-neuron current); where a number is quoted, the paper states the convention.

Symbol	Meaning
$N = 52$	internal/output neurons (ids 24–75)
$a = 52$	synaptic amplitude (current per spike = $a \cdot w$)
w	synaptic weight, $[0, 1]$
$t_e = 0.8$	M2 per-neuron excitatory budget
P_i	per-neuron protected (consolidated) mass
$p_{\max} = 0.75$	protected cap fraction; cap $p_{\max} t_e = 0.60$
$W_i = t_e - P_i$	working (unconsolidated) budget
$R_i = I_{p,i} / (I_{p,i} + I_{w,i})$	protected fraction of delivered input current (allocation gate)
$I_{W,i}$	working (unconsolidated) input-channel current delivered to i
$I_{\text{rec}}, I_{\text{inh}}$	recurrent excitatory / M6 inhibitory current
$v_i, v_{\text{th}} = 1, \tau_m = 20$	membrane potential, threshold, membrane time constant (ms)
u_i	V2.1 slow state; $+\beta/\text{spike}$, decay τ_s
$\beta = 0.0046875, \tau_s = 5000$	the e24 cell used in the CLLA era
$k_g = 8.0$	frozen recruitment-gain magnitude
$i_{\text{boost},i}$	recruitment-gain current (registration A; capped in B)
$\theta_{\text{perm}} = 0.05, w_{\text{c,perm}} = 0.02$	M3 permanence threshold / bootstrap weight

Conventions: “posts/n” is the fraction of the 52 neurons that fired at least once in a presentation window; “burst rate” is the presentation-window mean internal spike rate; “protected mean” is the mean weight over live consolidated synapses of a cohort; “drive end” is the snapshot at $t = 84,000$ ms (the e24 convention); per-presentation timing is 500 ms on / 1500 ms off. Where the records report network-total currents (e.g. 81.38 total vs. 1.565 per neuron at first C), the paper quotes the per-neuron value.

C Additional Quantitative Results

C.1 Key quantitative findings

C.2 Registration-B running details

Registration B’s matrix ran on the same nine arm configurations as registration A (run configs `c1la-rg8c-s{seed}-{order}`), identical identity gate (byte-identical flag-off telemetry against the committed corrected-fe baseline), identical endpoint bars, and the amended mechanism of Eq. (4) with the per-neuron threshold-completion clamp. Full run inventories and per-run telemetry are preserved under the `c1la-rg8-` and `c1la-rg8c-` run directories listed in the execution records [16, 21].

References

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Table 3: **Key quantitative findings and their primary sources.** All values are as recorded; per-neuron currents are marked.

Finding	Value	Source
A-trained afferent selectivity	+0.906; 49/52 neurons	[18]
Cross-run configuration cosine	0.119; LOPO run-identity 91%	[18]
Late-S1 cross-pattern cosine (E3/E3b)	0.673 / 0.740 (bar 0.60)	[?, 11]
Adjacent-state cosine under alternation	0.972 $\rightarrow$ 0.990	[18]
Structure vs. activity persistence (20 s silence)	cos 0.986–0.993 vs. u 3.5 $\times$, spikes 6.7 $\times$ decay	[18]
u plateau / Fano / core	$\bar{u} \rightarrow 1.32$; Fano ≈ 0 ; 5–6 neurons	[22]
Temporal capacity	< 50 ms ($D_{L-NF} = 0.0815$)	[19]
Blocked overwrite (recency)	BAC end A-mass 0.068 (from 0.208)	[18]
Decay as dominant eraser	decay-off preserves first cohort 6/6 (residual ~ 0.07 –0.09)	[19]
Unbounded CLLA runaway	6/12 aborts; P to 7.9 vs. cap 0.60	[6]
Bounded CLLA	12/12 complete; P pinned 0.6000; rates 3.5–6.5 Hz	[4]
CLLA capability	F1: il 3/3 PASS; blocked 6/6 FAIL; F3/4/5/7 PASS 12/12	[5]
Allocation rule primary	ratios 0.008–0.192 (bar 0.5); w_C 197 $\rightarrow$ 77 in block 1	[3]
Reserve C-cohort at onset	0.0 reserved (A 123.7)	[8]
fe time-to-permanence	2.4 s; 41 events (supply)	[12]
fec supply post-correction	61 events; 3.05/pres (C) vs. 13.2/pres (A)	[15]
fec protected end-mass	C 2.98 (mean 0.054) vs. A 22.87 (mean 0.096)	[15]
First-C post collapse	1,697 $\rightarrow$ 101 spikes (17 $\times$); C sustains 101–222	[20]
STDP net on C cohort (C block)	–21.63 over 1,961 events	[20]
First-C drive deficit	I_{aff} 1.57 vs. 11.98 (7.7 $\times$); I_{rec} 0.05 vs. 6.90	[14]
IL control first-C	I_{aff} 11.13, I_{rec} 7.05, posts 1.00; C perms 98 $\approx$ A 95	[14]
rg8 4b (pres-1–5 mean)	185.9–201.0 Hz (bar [100,200]); pres-1 391–449	[16]
rg8 first-C substrate	I_W 0.000–0.14 vs. 1.565 baseline	[16]
rg8c 4b aggregate	175.6–189.1 Hz; pres-1 378–440	[21]
rg8c primary endpoints	posts 0.00–0.19; perms 5–15; end-mean 0.000–0.092	[21]